

Ultimate 5m Technical Command Matrix

Master Specification & Feature Requirements Document

1. Core System Architecture

The system coordinates processing using an Orchestrator/CEO pattern, distributing calculations to sub-agents to ensure modularity, error tolerance, and thread protection.

ID	Feature Name	Core Functionality	Execution Protocol
A1	Multi-Agent CEO Orchestrator	Central system engine that handles signal generation	Sequentially gathers reports from all sub-agents (Tech, Inst, News) at candle close, compiles data, and executes final trade decisions based on consensus logic.
A2	System A: 5m Technical Engine	Technical chart indicator analysis (100 pts max score)	Computes indicators on closed 5m candles. Sends a calculated setup score to the CEO Orchestrator.
A3	System B: Institutional Edge	Market microstructure and order flow analysis	Monitors order flow metrics (Volume Delta, Cumulative Delta, DOM, Volume Profile, Options) and returns Neutral/Buy/Sell to CEO.
A4	News Intelligence Agent	Aggregates macro calendar releases and sentiment	Queries FRED, Finnhub, Forex Factory, and Reuters. Tracks fear indexes (VIX) and yield shifts to gate system execution.
A5	SEC Filing Agent	Insider activity and corporate change monitor	Scans SEC EDGAR every 10m (day) and 15m (night/weekend) for Form 4, SC 13D, and 8-K filings. Dispatches alerts independently.
A6	Position State Machine	Enforces trade direction boundaries and tracking	Reads/writes positions.json. Filters out redundant entry triggers. Allows exit commands only for open positions.

2. Data Connection Layer

ID	Feature Name	Data Target / Method	Protocol Specification
D1	IBKR WebSocket Feed	Real-Time Candles & Ticks	Subscribes to 5m bar stream and raw tick-by-tick streams per ticker. Relies on push events to run signal execution at exact candle closes.
D2	IBKR Historical Cache	90-Day Backtest History	Retrieves 90 days of 5m bars and caches data locally. Eliminates live queries during testing and guards against IP blocks.
D3	Rate Limit Manager	API Traffic Protection	Locks queries to 6 calls per 10-second window. Checks options chains on-demand only when System A score is above 50.
D4	Market Hours Gate	Session Verification Check	Restricts trading activity to US Regular Session (09:25 AM to 04:05 PM ET, Mon-Fri). Disables connections outside session hours.

3. System A Technical Indicator Parameters

ID	Indicator Name	Scoring Weight	Calculation Rules & Trade Rationale
S1	VWAP Bounce	20 Points	Price enters range [VWAP, VWAP + 0.15% ATR] and confirms with a bullish candle. Matches institutional execution zones.
S2	EMA 9/21 Ribbon	20 Points	EMA 9 crosses above EMA 21 within the last 2 bars. Measures short-term trend acceleration.
S3	MACD Slope	15 Points	MACD Histogram rises for 3 consecutive bars (Hist[0] > Hist[-1] > Hist[-2]). Confirms acceleration before a crossover.
S4	RSI Hidden Divergence	15 Points	Price prints higher low while RSI prints lower low. Captures seller exhaustion in consolidation zones.
S5	Volume Delta Confirm	15 Points	Current volume > 1.5x of the 20-period volume average. Confirms institutional backing on the breakout.
S6	Bollinger Band Squeeze	10 Points	Bollinger Bands contract inside Keltner Channels, then expand. Catches volatility cycle expansions.
S7	Order Block Proximity	5 Points	Price trades within 0.3% of the last down-close candle before a breakout swing. Identifies resting buy orders.

4. System B Institutional Order Flow Metrics

ID	Metric Name	Microstructure Target	Trading Utility
I1	Volume Delta	Net aggressive transaction volume	Categorizes trades at ask as buys and at bid as sells. Reveals real buying vs selling pressure.
I2	Cumulative Delta	Daily trend momentum tracker	Accumulates delta volume over the session. Confirms if trend control resides with buyers or sellers.
I3	DOM Imbalance	Order book depth comparison	Evaluates total size in top 10 bid/ask levels. Imbalance >= 3.0 flags major support blocks.
I4	Volume Profile	POC, VAH, and VAL calculations	Isolates high-volume price levels. Bounces at POC or VAH serve as low-risk entry zones.
I5	Options Chain Scan	Put/Call ratios and strike volume	Flags strikes where volume > 2.5x open interest. Highlights target zones defended by market makers.
I6	Dark Pool Approx	Off-exchange block cross monitor	Flags blocks (> 10k shares) executed at bid/ask midpoints. Identifies hidden institutional blocks.

5. Signal Output Definitions & Exit Strategy

ID	Signal Type	Trigger Criteria	Action / Target Plan
E1	Trend Continuation Buy	Tech Score >= 80 + Inst Engine is BUY	Executes full position size. Wider stop-loss (1.5x ATR). Projects Target 1 and Target 2.
E2	Momentum Scalp Buy	Tech Score 65-79 OR Tech >= 80 + Inst Neutral	Executes half position size. Tighter stop-loss (1.0x ATR). Target 2 disabled (auto-exits at Target 1).
E3	Conflict Info Alert	Tech Score >= 80 + Inst Engine is SELL	Skips execution. Dispatches conflict warning detailing logic split to Telegram and data logs.
E4	Emergency EXIT	Price Guardian stop-loss violation	Triggers immediate exit warning. Bypasses 5m candle close timer to protect capital.

Intelligent Exit Criteria Configurations

Trend Continuation Exits (2 of 5 conditions must match):

- 1. MACD histogram falls for 3 consecutive bars.
- 2. Volume Climax: volume > 2.5x average and close is in the bottom 30% of the candle's range.
- 3. RSI falls below 50 after crossing above 60.
- 4. Price closes below EMA 9 for 2 consecutive bars.
- 5. Locked trailing stop-loss is violated.

Momentum Scalp Exits (1 of 4 conditions must match):

- 1. MACD histogram falls for 2 consecutive bars.
- 2. Volume dries up to < 50% of the 20-bar average volume.
- 3. Price hits Target 1.
- 4. Tighter stop-loss (1.0x ATR) is violated.

6. Risk Controls, SEC Agent & Reporting

ID	Control / Agent	Triggers & Constraints	System Responses
R1	Price Guardian	5-second loop, stop breach / crash check	Fires emergency exit immediately if price violates trailing stop-loss or drops > 1.5% from close.
R2	Circuit Breakers	Loss targets: L1 (2 losses), L2 (3 losses), L3 (4 losses)	L1 sends warning. L2 penalizes all scores by -20 and disables scalps. L3 freezes all signals.
R3	VIX volatility gate	Fear index limits: VIX > 25, VIX > 35	VIX > 25 freezes scalps and cuts scores by -20. VIX > 35 halts the entire system.
R4	SEC EDGAR Scanner	Monitors Form 4, SC 13D, 8-K filings	Alerts Telegram immediately with transaction size, insider name, and links. Sweeps weekends at 10 AM.
R5	Reporting Modules	Interval triggers: EOD (4:05 PM), Weekly (Fri 4:30 PM)	EOD compiles P/L and plans tomorrow's calendars. Weekly analyzes indicators and suggests tunes.